

## SAFETY DATA SHEET

Creation Date 12-April-2023

Revision Date 02-April-2024

Revision Number 2

### 1. Identification

|                             |                                                                                                                           |
|-----------------------------|---------------------------------------------------------------------------------------------------------------------------|
| <b>Product Name</b>         | <b>Composite 5K, for volumetric one-component Karl Fischer titration in aldehydes and ketones, 5 mg H<sub>2</sub>O/mL</b> |
| <b>Cat No. :</b>            | <b>U90044</b>                                                                                                             |
| <b>Synonyms</b>             | No information available                                                                                                  |
| <b>Recommended Use</b>      | Laboratory chemicals.                                                                                                     |
| <b>Uses advised against</b> | Food, drug, pesticide or biocidal product use.                                                                            |

#### Details of the supplier of the safety data sheet

##### Company

##### **Importer/Distributor**

Fisher Scientific  
112 Colonnade Road,  
Ottawa, ON K2E 7L6,  
Canada  
Tel: 1-800-234-7437

##### **Emergency Telephone Number**

For information **US** call: 001-800-227-6701 / **Europe** call: +32 14 57 52 11

Emergency Number **US**:001-201-796-7100 / **Europe**: +32 14 57 52 99

**CHEMTREC** Tel. No. **US**:001-800-424-9300 / **Europe**:001-703-527-3887

### 2. Hazard(s) identification

#### Classification

##### **WHMIS 2015 Classification**

Classified as hazardous under the Hazardous Products Regulations (SOR/2015-17)

|                                                             |              |
|-------------------------------------------------------------|--------------|
| <b>Flammable liquids</b>                                    | Category 3   |
| <b>Acute oral toxicity</b>                                  | Category 4   |
| <b>Acute dermal toxicity</b>                                | Category 4   |
| <b>Acute Inhalation Toxicity</b>                            | Category 4   |
| <b>Skin Corrosion/Irritation</b>                            | Category 1 B |
| <b>Serious Eye Damage/Eye Irritation</b>                    | Category 1   |
| <b>Reproductive Toxicity</b>                                | Category 1B  |
| <b>Specific target organ toxicity (single exposure)</b>     | Category 1   |
| Target Organs - Respiratory system.                         |              |
| <b>Specific target organ toxicity - (repeated exposure)</b> | Category 2   |
| <b>Physical Hazards Not Otherwise Classified</b>            | Category 1   |
| May form explosive peroxides                                |              |

#### Label Elements

**Signal Word**

Danger

**Hazard Statements**

Flammable liquid and vapor  
Harmful if swallowed, in contact with skin or if inhaled  
Causes severe skin burns and eye damage  
May cause respiratory irritation  
May damage the unborn child. Suspected of damaging fertility  
May damage the unborn child  
May damage fertility. May damage the unborn child  
Causes damage to organs  
May cause damage to organs through prolonged or repeated exposure  
May form explosive peroxides



**Precautionary Statements**

**Prevention**

Keep container tightly closed  
Obtain special instructions before use  
Do not handle until all safety precautions have been read and understood  
Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking  
Ground/bond container and receiving equipment  
Use non-sparking tools  
Take action to prevent static discharges  
Do not breathe dust/fumes/gas/mist/vapours/spray  
Wash face, hands and any exposed skin thoroughly after handling  
Do not eat, drink or smoke when using this product  
Use only outdoors or in a well-ventilated area  
Wear protective gloves/protective clothing/eye protection/face protection

**Response**

IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water/ shower  
IF INHALED: Remove person to fresh air and keep comfortable for breathing  
IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing  
Immediately call a POISON CENTER/doctor  
Rinse mouth  
Do NOT induce vomiting  
Wash contaminated clothing before reuse  
In case of fire: Use dry sand, dry chemical or alcohol-resistant foam to extinguish

**Storage**

Store locked up  
Store in a well-ventilated place. Keep container tightly closed

**Disposal**

Dispose of contents/container to an approved waste disposal plant

### 3. Composition/Information on Ingredients

| Component                     | CAS-No    | Weight % |
|-------------------------------|-----------|----------|
| 2-Methoxyethanol              | 109-86-4  | 61       |
| Ethylene glycol diethyl ether | 629-14-1  | 12       |
| Iodine                        | 7553-56-2 | 11       |
| 1-Imidazole                   | 288-32-4  | 10       |
| Sulfur dioxide                | 7446-09-5 | 6        |

#### 4. First-aid measures

|                                        |                                                                                                                                                                                                                                                                                                                                                                                      |
|----------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>General Advice</b>                  | Show this safety data sheet to the doctor in attendance. Immediate medical attention is required.                                                                                                                                                                                                                                                                                    |
| <b>Eye Contact</b>                     | Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Immediate medical attention is required.                                                                                                                                                                                                                                                    |
| <b>Skin Contact</b>                    | Wash off immediately with plenty of water for at least 15 minutes. Remove and wash contaminated clothing and gloves, including the inside, before re-use. Call a physician immediately.                                                                                                                                                                                              |
| <b>Inhalation</b>                      | If not breathing, give artificial respiration. Remove from exposure, lie down. Do not use mouth-to-mouth method if victim ingested or inhaled the substance; give artificial respiration with the aid of a pocket mask equipped with a one-way valve or other proper respiratory medical device. Call a physician immediately.                                                       |
| <b>Ingestion</b>                       | Do NOT induce vomiting. Clean mouth with water. Never give anything by mouth to an unconscious person. Call a physician immediately.                                                                                                                                                                                                                                                 |
| <b>Most important symptoms/effects</b> | Causes burns by all exposure routes. Product is a corrosive material. Use of gastric lavage or emesis is contraindicated. Possible perforation of stomach or esophagus should be investigated: Ingestion causes severe swelling, severe damage to the delicate tissue and danger of perforation: Symptoms of overexposure may be headache, dizziness, tiredness, nausea and vomiting |
| <b>Notes to Physician</b>              | Treat symptomatically                                                                                                                                                                                                                                                                                                                                                                |

#### 5. Fire-fighting measures

|                                         |                                                                                                                     |
|-----------------------------------------|---------------------------------------------------------------------------------------------------------------------|
| <b>Suitable Extinguishing Media</b>     | CO <sub>2</sub> , dry chemical, dry sand, alcohol-resistant foam. Water mist may be used to cool closed containers. |
| <b>Unsuitable Extinguishing Media</b>   | No information available                                                                                            |
| <b>Flash Point</b>                      | No information available                                                                                            |
| <b>Method -</b>                         | No information available                                                                                            |
| <b>Autoignition Temperature</b>         | No information available                                                                                            |
| <b>Explosion Limits</b>                 |                                                                                                                     |
| <b>Upper</b>                            | No data available                                                                                                   |
| <b>Lower</b>                            | No data available                                                                                                   |
| <b>Sensitivity to Mechanical Impact</b> | No information available                                                                                            |
| <b>Sensitivity to Static Discharge</b>  | No information available                                                                                            |

#### Specific Hazards Arising from the Chemical

Thermal decomposition can lead to release of irritating gases and vapors. The product causes burns of eyes, skin and mucous membranes. Flammable. Containers may explode when heated. Vapors may form explosive mixtures with air. Vapors may travel to source of ignition and flash back.

### Hazardous Combustion Products

Carbon oxides. Nitrogen oxides (NO<sub>x</sub>). Sulfur oxides. Hydrogen iodide.

### Protective Equipment and Precautions for Firefighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear. Thermal decomposition can lead to release of irritating gases and vapors.

### NFPA

**Health**  
3

**Flammability**  
4

**Instability**  
0

**Physical hazards**  
N/A

## 6. Accidental release measures

### Personal Precautions

Ensure adequate ventilation. Use personal protective equipment as required. Evacuate personnel to safe areas. Keep people away from and upwind of spill/leak. Remove all sources of ignition. Take precautionary measures against static discharges.

### Environmental Precautions

Do not flush into surface water or sanitary sewer system.

### Methods for Containment and Clean Up

Soak up with inert absorbent material. Keep in suitable, closed containers for disposal. Remove all sources of ignition. Use spark-proof tools and explosion-proof equipment.

## 7. Handling and storage

### Handling

Wear personal protective equipment/face protection. Do not get in eyes, on skin, or on clothing. Use only under a chemical fume hood. Do not breathe mist/vapors/spray. Do not ingest. If swallowed then seek immediate medical assistance. If peroxide formation is suspected, do not open or move container. Keep away from open flames, hot surfaces and sources of ignition. Use only non-sparking tools. Take precautionary measures against static discharges.

### Storage.

Corrosives area. Keep containers tightly closed in a dry, cool and well-ventilated place. Containers should be dated when opened and tested periodically for the presence of peroxides. Should crystals form in a peroxidizable liquid, peroxidation may have occurred and the product should be considered extremely dangerous. In this instance, the container should only be opened remotely by professionals. Keep away from heat, sparks and flame.

## 8. Exposure controls / personal protection

### Exposure Guidelines

| Component        | Alberta                                                                               | British Columbia          | Ontario TWA/EV                                                                          | Quebec                                                                                | ACGIH TLV                      | OSHA PEL                                                                                                                       | NIOSH                                                                                                |
|------------------|---------------------------------------------------------------------------------------|---------------------------|-----------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|--------------------------------|--------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|
| 2-Methoxyethanol | TWA: 0.1 ppm<br>TWA: 0.3 mg/m <sup>3</sup><br>Skin                                    | TWA: 0.1 ppm<br>Skin      | TWA: 0.1 ppm<br>Skin                                                                    | TWA: 5 ppm<br>TWA: 16 mg/m <sup>3</sup><br>Skin                                       | TWA: 0.1 ppm<br>Skin           | (Vacated) TWA: 25 ppm<br>(Vacated) TWA: 80 mg/m <sup>3</sup><br>Skin<br>TWA: 25 ppm<br>TWA: 80 mg/m <sup>3</sup>               | IDLH: 200 ppm<br>TWA: 0.1 ppm<br>TWA: 0.3 mg/m <sup>3</sup>                                          |
| Iodine           | Ceiling: 0.1 ppm<br>Ceiling: 1 mg/m <sup>3</sup>                                      | Ceiling: 0.1 ppm          | TWA: 0.01 ppm<br>STEL: 0.1 ppm                                                          | Ceiling: 0.1 ppm<br>Ceiling: 1.0 mg/m <sup>3</sup>                                    | TWA: 0.01 ppm<br>STEL: 0.1 ppm | Ceiling: 0.1 ppm<br>Ceiling: 1 mg/m <sup>3</sup><br>(Vacated)<br>Ceiling: 0.1 ppm<br>(Vacated)<br>Ceiling: 1 mg/m <sup>3</sup> | IDLH: 2 ppm<br>Ceiling: 0.1 ppm<br>Ceiling: 1 mg/m <sup>3</sup>                                      |
| Sulfur dioxide   | TWA: 2 ppm<br>TWA: 5.2 mg/m <sup>3</sup><br>STEL: 5 ppm<br>STEL: 13 mg/m <sup>3</sup> | TWA: 2 ppm<br>STEL: 5 ppm | TWA: 2 ppm<br>TWA: 5.2 mg/m <sup>3</sup><br>STEL: 5 ppm<br>STEL: 10.4 mg/m <sup>3</sup> | TWA: 2 ppm<br>TWA: 5.2 mg/m <sup>3</sup><br>STEL: 5 ppm<br>STEL: 13 mg/m <sup>3</sup> | STEL: 0.25 ppm                 | (Vacated) TWA: 2 ppm<br>(Vacated) TWA: 5 mg/m <sup>3</sup><br>(Vacated) STEL: 5 ppm                                            | IDLH: 100 ppm<br>TWA: 2 ppm<br>TWA: 5 mg/m <sup>3</sup><br>STEL: 5 ppm<br>STEL: 13 mg/m <sup>3</sup> |

|  |  |  |  |  |  |                                                                                    |  |
|--|--|--|--|--|--|------------------------------------------------------------------------------------|--|
|  |  |  |  |  |  | (Vacated) STEL:<br>15 mg/m <sup>3</sup><br>TWA: 5 ppm<br>TWA: 13 mg/m <sup>3</sup> |  |
|--|--|--|--|--|--|------------------------------------------------------------------------------------|--|

**Legend**

ACGIH - American Conference of Governmental Industrial Hygienists

OSHA - Occupational Safety and Health Administration

NIOSH: NIOSH - National Institute for Occupational Safety and Health

**Engineering Measures**

Ensure that eyewash stations and safety showers are close to the workstation location.

Ensure adequate ventilation, especially in confined areas. Use explosion-proof electrical/ventilating/lighting/equipment.

Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source

**Personal protective equipment**

**Eye Protection**

Goggles

**Hand Protection**

Wear appropriate protective gloves and clothing to prevent skin exposure.

| Glove material | Breakthrough time | Glove thickness | Glove comments         |
|----------------|-------------------|-----------------|------------------------|
| Nitrile rubber | See manufacturers | -               | Splash protection only |
| Neoprene       | recommendations   |                 |                        |
| Natural rubber |                   |                 |                        |
| PVC            |                   |                 |                        |

Inspect gloves before use. observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information) gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion. gloves with care avoiding skin contamination.

**Respiratory Protection**

When workers are facing concentrations above the exposure limit they must use appropriate certified respirators. Follow the OSHA respirator regulations found in 29 CFR 1910.134 or European Standard EN 149. Use a NIOSH/MSHA or European Standard EN 149 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced.

To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained properly

**Recommended Filter type:** Organic gases and vapours filter Type A Brown conforming to EN14387

When RPE is used a face piece Fit Test should be conducted

**Environmental exposure controls**

Prevent product from entering drains. Do not allow material to contaminate ground water system. Local authorities should be advised if significant spillages cannot be contained.

**Hygiene Measures**

Handle in accordance with good industrial hygiene and safety practice. Keep away from food, drink and animal feeding stuffs. Do not eat, drink or smoke when using this product. Remove and wash contaminated clothing and gloves, including the inside, before re-use. Wash hands before breaks and after work.

## 9. Physical and chemical properties

**Physical State**

Liquid

**Appearance**

No information available

**Odor**

No information available

**Odor Threshold**

No information available

**pH**

Not applicable

|                                        |                          |
|----------------------------------------|--------------------------|
| Melting Point/Range                    | No data available        |
| Boiling Point/Range                    | No information available |
| Flash Point                            | No information available |
| Evaporation Rate                       | No information available |
| Flammability (solid,gas)               | Not applicable           |
| Flammability or explosive limits       |                          |
| Upper                                  | No data available        |
| Lower                                  | No data available        |
| Vapor Pressure                         | No information available |
| Vapor Density                          | No information available |
| Density                                | 0.93                     |
| Specific Gravity                       | No information available |
| Solubility                             | No information available |
| Partition coefficient; n-octanol/water | No data available        |
| Autoignition Temperature               | No information available |
| Decomposition Temperature              | No information available |
| Viscosity                              | No information available |

## 10. Stability and reactivity

|                                  |                                                                      |
|----------------------------------|----------------------------------------------------------------------|
| Reactive Hazard                  | None known, based on information available                           |
| Stability                        | Stable under normal conditions.                                      |
| Conditions to Avoid              | Keep away from open flames, hot surfaces and sources of ignition.    |
| Incompatible Materials           | Strong oxidizing agents                                              |
| Hazardous Decomposition Products | Carbon oxides, Nitrogen oxides (NOx), Sulfur oxides, Hydrogen iodide |
| Hazardous Polymerization         | Hazardous polymerization does not occur.                             |
| Hazardous Reactions              | None under normal processing.                                        |

## 11. Toxicological information

### Acute Toxicity

#### Product Information

**Oral LD50** Based on ATE data, the classification criteria are not met. ATE > 2000 mg/kg. Category 4.  
ATE = 300 - 2000 mg/kg.

**Dermal LD50** Based on ATE data, the classification criteria are not met. ATE > 2000 mg/kg. Category 4.  
ATE = 1000 - 2000 mg/kg.

**Vapor LC50** Category 4. ATE = 10 - 20 mg/l.

#### Component Information

| Component                     | LD50 Oral                 | LD50 Dermal                  | LC50 Inhalation                    |
|-------------------------------|---------------------------|------------------------------|------------------------------------|
| 2-Methoxyethanol              | LD50 = 2370 mg/kg ( Rat ) | LD50 = 1280 mg/kg ( Rabbit ) | LC50 = 1478 ppm ( Rat ) 7 h        |
| Ethylene glycol diethyl ether | LD50 = 4390 mg/kg ( Rat ) | LD50 = 8 mL/kg ( Rabbit )    | Not listed                         |
| Iodine                        | 315 mg/kg ( Rat )         | 1425 mg/kg ( Rabbit )        | 4.588 mg/L 4h ( Rat )              |
| 1-Imidazole                   | 970 mg/kg ( Rat )         | -                            | -                                  |
| Sulfur dioxide                | Not listed                | Not listed                   | Per CGA P-20: 2500 ppm/1hr ( Rat ) |

**Toxicologically Synergistic Products** No information available

### Delayed and immediate effects as well as chronic effects from short and long-term exposure

**Irritation** No information available

**Sensitization** No information available

**Carcinogenicity** The table below indicates whether each agency has listed any ingredient as a carcinogen.

| Component                     | CAS-No    | IARC       | NTP        | ACGIH      | OSHA       | Mexico     |
|-------------------------------|-----------|------------|------------|------------|------------|------------|
| 2-Methoxyethanol              | 109-86-4  | Not listed | Not listed | Not listed | Not listed | Not listed |
| Ethylene glycol diethyl ether | 629-14-1  | Not listed | Not listed | Not listed | Not listed | Not listed |
| Iodine                        | 7553-56-2 | Not listed | Not listed | Not listed | Not listed | Not listed |
| 1-Imidazole                   | 288-32-4  | Not listed | Not listed | Not listed | Not listed | Not listed |
| Sulfur dioxide                | 7446-09-5 | Not listed | Not listed | Not listed | Not listed | Not listed |

**Mutagenic Effects** No information available

**Reproductive Effects** No information available.

**Developmental Effects** No information available.

**Teratogenicity** No information available.

**STOT - single exposure** Respiratory system

**STOT - repeated exposure** None known

**Aspiration hazard** No information available

**Symptoms / effects, both acute and delayed** Product is a corrosive material. Use of gastric lavage or emesis is contraindicated. Possible perforation of stomach or esophagus should be investigated: Ingestion causes severe swelling, severe damage to the delicate tissue and danger of perforation: Symptoms of overexposure may be headache, dizziness, tiredness, nausea and vomiting

**Endocrine Disruptor Information** No information available

**Other Adverse Effects** The toxicological properties have not been fully investigated.

## 12. Ecological information

### Ecotoxicity

The product contains following substances which are hazardous for the environment. Contains a substance which is: Very toxic to aquatic organisms.

| Component        | Freshwater Algae                                                                                  | Freshwater Fish                                                                                                                                                     | Microtox                                                                                      | Water Flea                              |
|------------------|---------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------|-----------------------------------------|
| 2-Methoxyethanol | Not listed                                                                                        | LC50: = 9650 mg/L, 96h static (Lepomis macrochirus)<br>LC50: = 16000 mg/L, 96h static (Oncorhynchus mykiss)<br>LC50: = 10000 mg/L, 96h static (Lepomis macrochirus) | Not listed                                                                                    | Not listed                              |
| Iodine           | -                                                                                                 | Oncorhynchus mykiss: LC50 = 1,7 mg/l/96 h                                                                                                                           | -                                                                                             | EC50 = 0,2 mg/l/48 h                    |
| 1-Imidazole      | EC50: = 82 mg/L, 96h (Desmodesmus subspicatus)<br>EC50: = 130 mg/L, 72h (Desmodesmus subspicatus) | Not listed                                                                                                                                                          | = 1200 mg/L EC50 Pseudomonas putida 17 h<br>= 231 mg/L EC50 Photobacterium phosphoreum 30 min | EC50: = 341.5 mg/L, 48h (Daphnia magna) |

**Persistence and Degradability** No information available

**Bioaccumulation/ Accumulation** No information available.

Mobility No information available.

| Component        | log Pow |
|------------------|---------|
| 2-Methoxyethanol | -0.77   |
| Iodine           | 2.49    |
| 1-Imidazole      | -0.02   |

### 13. Disposal considerations

**Waste Disposal Methods** Chemical waste generators must determine whether a discarded chemical is classified as a hazardous waste. Chemical waste generators must also consult local, regional, and national hazardous waste regulations to ensure complete and accurate classification.

### 14. Transport information

#### DOT

UN-No UN2924  
Proper Shipping Name FLAMMABLE LIQUIDS, CORROSIVE, N.O.S.  
Technical Name 2-Methoxyethanol, 1-Imidazole  
Hazard Class 3  
Subsidiary Hazard Class 8  
Packing Group III

#### TDG

UN-No UN2924  
Proper Shipping Name FLAMMABLE LIQUID, CORROSIVE, N.O.S.  
Hazard Class 3  
Subsidiary Hazard Class 8  
Packing Group III

#### IATA

UN-No UN2924  
Proper Shipping Name FLAMMABLE LIQUID, CORROSIVE, N.O.S.  
Hazard Class 3  
Subsidiary Hazard Class 8  
Packing Group III

#### IMDG/IMO

UN-No UN2924  
Proper Shipping Name FLAMMABLE LIQUID, CORROSIVE, N.O.S.  
Hazard Class 3  
Subsidiary Hazard Class 8  
Packing Group III

### 15. Regulatory information

All of the components in the product are on the following inventory lists: China X = listed Australia U.S.A. (TSCA) Canada (DSL/NDL) Europe (EINECS/ELINCS/NLP) Australia (AICS) Korea (KECL) China (IECSC) Japan (ENCS) Philippines (PICCS) Taiwan (TCSI) Japan (ISHL) New Zealand (NZIoC) Japan (ISHL)

#### International Inventories

| Component                     | CAS-No    | DSL | NDL | TSCA | TSCA Inventory notification - Active-Inactive | EINECS    | ELINCS | NLP |
|-------------------------------|-----------|-----|-----|------|-----------------------------------------------|-----------|--------|-----|
| 2-Methoxyethanol              | 109-86-4  | X   | -   | X    | ACTIVE                                        | 203-713-7 | -      | -   |
| Ethylene glycol diethyl ether | 629-14-1  | -   | -   | X    | ACTIVE                                        | 211-076-1 | -      | -   |
| Iodine                        | 7553-56-2 | X   | -   | X    | ACTIVE                                        | 231-442-4 | -      | -   |
| 1-Imidazole                   | 288-32-4  | X   | -   | X    | ACTIVE                                        | 206-019-2 | -      | -   |
| Sulfur dioxide                | 7446-09-5 | X   | -   | X    | ACTIVE                                        | 231-195-2 | -      | -   |

| Component | CAS-No | IECSC | KECL | ENCS | ISHL | TCSI | AICS | NZIoC | PICCS |
|-----------|--------|-------|------|------|------|------|------|-------|-------|
|-----------|--------|-------|------|------|------|------|------|-------|-------|



**Composite 5K, for volumetric one-component Karl  
Fischer titration in aldehydes and ketones, 5 mg  
H<sub>2</sub>O/mL**

Revision Date 02-April-2024

|                               |           |   |          |   |   |   |   |   |   |
|-------------------------------|-----------|---|----------|---|---|---|---|---|---|
| 2-Methoxyethanol              | 109-86-4  | X | KE-23272 | X | X | X | X | X | X |
| Ethylene glycol diethyl ether | 629-14-1  | X | KE-10343 | X | X | X | X | X | X |
| Iodine                        | 7553-56-2 | X | KE-21023 | X | - | X | X | X | X |
| 1-Imidazole                   | 288-32-4  | X | KE-20937 | X | X | X | X | X | X |
| Sulfur dioxide                | 7446-09-5 | X | KE-32567 | X | X | X | X | X | X |

**Legend:**

X - Listed '-' - Not Listed

KECL - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)

DSL/NDL - Canadian Domestic Substances List/Non-Domestic Substances List

TSCA - United States Toxic Substances Control Act Section 8(b) Inventory

EINECS/ELINCS - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

IECSC - Chinese Inventory of Existing Chemical Substances

KECL - Korean Existing and Evaluated Chemical Substances

ENCS - Japanese Existing and New Chemical Substances

AICS - Australian Inventory of Chemical Substances

PICCS - Philippines Inventory of Chemicals and Chemical Substances

**Canada**

SDS in compliance with provisions of information as set out in Canadian Standard - Part 4, Schedule 1 and 2 of the Hazardous Products Regulations (HPR) and meets the requirements of the HPR (Paragraph 13(1)(a) of the Hazardous Products Act (HPA)).

| Component                     | Canada - National Pollutant Release Inventory (NPRI) | Canadian Environmental Protection Agency (CEPA) - List of Toxic Substances | Canada's Chemicals Management Plan (CEPA) |
|-------------------------------|------------------------------------------------------|----------------------------------------------------------------------------|-------------------------------------------|
| 2-Methoxyethanol              | Part 1, Group A Substance Part 4 Substance           | Schedule I                                                                 |                                           |
| Ethylene glycol diethyl ether | Part 4 Substance                                     |                                                                            |                                           |
| Sulfur dioxide                | Part 4 Substance                                     | Schedule I                                                                 |                                           |

**Legend**

NPRI - National Pollutant Release Inventory

**Other International Regulations**

**Authorisation/Restrictions according to EU REACH**

| Component                     | REACH (1907/2006) - Annex XIV - Substances Subject to Authorization | REACH (1907/2006) - Annex XVII - Restrictions on Certain Dangerous Substances                                                      | REACH Regulation (EC 1907/2006) article 59 - Candidate List of Substances of Very High Concern (SVHC) |
|-------------------------------|---------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| 2-Methoxyethanol              | -                                                                   | Use restricted. See item 30. (see link for restriction details)<br>Use restricted. See item 75. (see link for restriction details) | SVHC Candidate list - 203-713-7 - Toxic for reproduction, Article 57c                                 |
| Ethylene glycol diethyl ether | -                                                                   | Use restricted. See item 30. (see link for restriction details)<br>Use restricted. See item 75. (see link for restriction details) | SVHC Candidate list - Toxic for reproduction (Article 57 c)                                           |
| Iodine                        | -                                                                   | Use restricted. See item 75. (see link for restriction details)                                                                    | -                                                                                                     |
| 1-Imidazole                   | -                                                                   | Use restricted. See item 30. (see link for restriction details)<br>Use restricted. See item 75. (see link for restriction details) | -                                                                                                     |
| Sulfur dioxide                | -                                                                   | Use restricted. See item 75. (see link for restriction details)                                                                    | -                                                                                                     |

After the sunset date the use of this substance requires either an authorization or can only be used for exempted uses, e.g. use in scientific research and development which includes routine analytics or use as intermediate.

**REACH links**

<https://echa.europa.eu/authorisation-list>

<https://echa.europa.eu/substances-restricted-under-reach>  
<https://echa.europa.eu/candidate-list-table>

**Safety, health and environmental regulations/legislation specific for the substance or mixture**

| Component                     | CAS-No    | OECD HPV       | Persistent Organic Pollutant | Ozone Depletion Potential | Restriction of Hazardous Substances (RoHS) |
|-------------------------------|-----------|----------------|------------------------------|---------------------------|--------------------------------------------|
| 2-Methoxyethanol              | 109-86-4  | Listed         | Not applicable               | Not applicable            | Not applicable                             |
| Ethylene glycol diethyl ether | 629-14-1  | Not applicable | Not applicable               | Not applicable            | Not applicable                             |
| Iodine                        | 7553-56-2 | Listed         | Not applicable               | Not applicable            | Not applicable                             |
| 1-Imidazole                   | 288-32-4  | Listed         | Not applicable               | Not applicable            | Not applicable                             |
| Sulfur dioxide                | 7446-09-5 | Listed         | Not applicable               | Not applicable            | Not applicable                             |

| Component                     | CAS-No    | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements | Rotterdam Convention (PIC) | Basel Convention (Hazardous Waste) |
|-------------------------------|-----------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|----------------------------|------------------------------------|
| 2-Methoxyethanol              | 109-86-4  | Not applicable                                                                            | Not applicable                                                                           | Not applicable             | Not applicable                     |
| Ethylene glycol diethyl ether | 629-14-1  | Not applicable                                                                            | Not applicable                                                                           | Not applicable             | Not applicable                     |
| Iodine                        | 7553-56-2 | Not applicable                                                                            | Not applicable                                                                           | Not applicable             | Not applicable                     |
| 1-Imidazole                   | 288-32-4  | Not applicable                                                                            | Not applicable                                                                           | Not applicable             | Not applicable                     |
| Sulfur dioxide                | 7446-09-5 | Not applicable                                                                            | Not applicable                                                                           | Not applicable             | Not applicable                     |

## 16. Other information

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New emergency telephone response service provider.

**Disclaimer**

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**End of SDS**